

TECHNOLOGY CONSULTING

IN THE GLOBAL COMMUNITY

Final Consulting Report

Rwanda ASYV

Kajal Mehta, Jiatong Li

August 2025

Carnegie Mellon University



Alumni Management System

Executive Summary

Student Consultants, Kajal Mehta & Jiatong Li
Community Partners, Stella Wayanzuvuko & Joseph Ngendahimana

I. About the Organization

Agahozo-Shalom Youth Village (ASYV) is a nonprofit in Rwanda founded in 2008 by Anne Heyman and inspired by the Israeli youth village model. As articulated in their mission statement:

“Through healing, education, and love, the Agahozo-Shalom Youth Village empowers orphaned and vulnerable Rwandan youth to build lives of dignity and contribute to a better world”

ASYV provides a holistic, residential community where students live in family-style homes, receive a strong academic and extracurricular education, and gain the tools to become leaders in their communities.

ASYV is currently expanding its digital infrastructure. The refactored Alumni Management System (AMS), built with Django, PostgreSQL, and React, aims to better track alumni outcomes and connect the ASYV community. Improving this system—particularly its responsiveness to different interfaces, front-end visualization and data accuracy—will support program evaluation, fundraising, and long-term impact. This work aligns with ASYV’s broader vision of creating an integrated digital platform to manage school and village life.

II. Making the Alumni Management System Responsive

ASYV’s Alumni Management System (AMS) previously lacked mobile accessibility, limiting alumni engagement—especially among users who only access the internet via smartphones. To address this, we led a full front-end redesign of the AMS interface to make it responsive across desktop, tablet, and mobile devices.

Outputs and Outcomes:

- Fully responsive layouts for all screen sizes, including simplified one-column views for small devices
- Mobile navigation overhaul with a hamburger menu replacing the sidebar
- Restructured components (e.g., alumni profiles, directory cards) for clarity and accessibility
- Live testing with alumni on four phone models to ensure usability

Risks and Limitations:

- Some interface features remain unintuitive and could benefit from improved UX design
- Administrative tools are less optimized for mobile, though primarily used on desktops

III. Redesigning the Database and Website Interface

The original AMS database and interface did not meet the long-term needs of ASYV's Career Resource Center (CRC) or broader organizational goals for tracking student and alumni outcomes. We redesigned the backend database schema and aligned it with new front-end tools to enable scalable, filterable, and actionable data tracking.

Outputs and Outcomes:

- A modular, normalized database design to track current students, alumni, and staff
- Admin tools for uploading and editing records, managing accounts, and assigning roles
- Filterable student and alumni directories by multiple criteria (e.g., grade, family, college)
- A CRC analytics dashboard to visualize outcomes in real time
- A redesigned alumni dashboard previewing jobs, events, and education opportunities

Risks and Limitations:

- Data remains incomplete or inconsistent; staff-led cleanup is ongoing
- Several usability enhancements can help streamline interaction

IV. Additional Recommendations

We propose some features to further strengthen the AMS and support alumni engagement:

- Support Request Logging: Allow alumni to request help from CRC through the platform and track follow-ups, reducing WhatsApp reliance
- Event RSVPs: Complete the RSVP system to log alumni interest and help staff manage events more effectively
- Resume Uploads: Enable alumni to upload resumes to their profiles for easier CRC access and support
- Dropdowns and Data Validation: Replace more free-text fields with dropdowns and add validation to ensure data consistency
- Alumni Connection Platform: Support a student-led initiative to build a messaging system for alumni networking, addressing a strongly expressed user need

Consulting Partner

Stella Wayianzuvuko
stella@asyv.org

Joseph Ngendahimana
joseph@asyv.org

Agahozo-Shalom Youth Village
Rwamagana District,
Rubona Sector, Rwanda
<http://www.asyv.org>

About the Consultants

Jiatong Li
jiatong4@andrew.cmu.edu

Kajal Mehta
kajalm@andrew.cmu.edu

Jiatong and Kajal both recently graduated with a Bachelor's in Information Systems
Jiatong will start her masters and Kajal will begin work this fall.

Alumni Management System Final Report

Student Consultants, Kajal Mehta & Jiatong Li
Community Partners, Stella Wayanzuvuko & Joseph Ngendahimana

I. About the Organization

Agahozo-Shalom Youth Village (ASYV) is a nonprofit organization in Rwanda that provides a nurturing, residential community for orphaned and vulnerable youth. Established in 2008 by Anne Heyman, a South African-American lawyer, ASYV was inspired by the Israeli youth village model, Yemin Orde, which was created to care for orphans of the Holocaust. The name "Agahozo-Shalom" combines the Kinyarwanda word "agahozo," meaning "a place where tears are dried," and the Hebrew word "shalom," meaning "peace," reflecting the village's mission to offer healing and tranquility to its residents, as articulated in their mission statement:

“Through healing, education, and love, the Agahozo-Shalom Youth Village empowers orphaned and vulnerable Rwandan youth to build lives of dignity and contribute to a better world”

The village employs a holistic approach centered around four pillars:

1. Parental Wholeness: Students are integrated into family-style homes led by a "Mama," fostering a sense of belonging and emotional support
2. Formal Education: Aligned with the Rwandan National Curriculum, ASYV offers rigorous academic programs, including science and humanities tracks, with instruction in English.
3. Health and Wellness: Comprehensive healthcare services, nutritious meals, and clean water are provided, alongside mental health support from social workers.
4. Life Enrichment Applied Programs (LEAP): Students engage in extracurricular activities such as sports, arts, music, and community service, promoting personal growth and leadership skills.

ASYV has supported over 1,900 youth, with more than 1,400 alumni pursuing diverse paths as entrepreneurs, educators, and professionals both within Rwanda and internationally. In addition to its core programs, ASYV launched the Educational Resilience Program (ERP) in 2021, aiming to train teachers across Rwanda. By the end of 2024, ERP is projected to impact up to 125,000 children nationwide.

Facilities

The Agahozo-Shalom Youth Village (ASYV) is a 150-acre campus located 40 kilometers east of Kigali, Rwanda, set among rolling hills and farmland. The site is securely fenced and monitored 24/7.

Facilities include a school with 22 classrooms and computer labs, administrative offices, student residences (housing 20–24 students each with a family-style caregiver), a dining hall, canteen, sports fields, clubhouses, a farm, solar field, guest houses, a visitor center, and a Health & Wellness Clinic.

The computer labs are outfitted with HP desktops and Dell laptops running Windows. Wireless internet is available but limited by a 16 Mbps bandwidth, often resulting in slow connectivity. Electricity is delivered at 220V and can fluctuate, which, along with limited bandwidth, poses challenges for reliable computer use.

Programs

The Agahozo-Shalom Youth Village (ASYV) offers two integrated educational programs: formal and informal education, structured around a three-term academic year. The formal program includes an enrichment year (EY) focused on English, Digital Literacy, Mathematics, and Science, followed by upper-secondary levels (Senior 4–6), where students select from the following academic combinations in addition to taking English, Entrepreneurship, and IT courses:

- Physics, Computer science, Math
- Math, Economics, Computers
- Math, Economics, Geography
- History, Economics, Geography (no longer applicable)
- History, Geography, Literature
- Physics, Biology, Chemistry

Informal education supplements this with arts, sports, and service learning. The art component covers a wide range of mediums such as digital media, recording, photography, and filmmaking. The service aspect, known as "Tikkun Olam," meaning "repairing the world," involves students engaging in activities like teaching English in nearby schools, constructing houses for underprivileged villages, assisting in clinics, or receiving training related to the One Laptop Per Child initiative. Each evening, there is designated family time that includes debates, discussions, and engaging activities. On Friday evenings, the entire village gathers at the amphitheater to showcase their diverse talents. Saturdays are often spent volunteering on the farm or enjoying movies, while Sundays involve attending church services and carrying out household cleaning tasks.

Technology plays a central role in both academics and operations. Computer labs support digital literacy and subject-specific learning, while tools like Google Classroom streamline teaching tasks like grading, and resource sharing.

Additionally, the Educational Resilience Program (ERP) trains Rwandan educators in tech-enabled, student-centered teaching and life skills education. Within ASYV itself, there has also been an initiative to train teachers on AI so they are better equipped to integrate AI into student learning.

Staff

Throughout the project we worked side by side with **Joseph Ngendahimana** (*School Learning Management System Administrator*). Joseph joined ASYV in 2019 and is the IT technician and used to be a computer science teacher here. We also collaborated with Fellows who audited the past Alumni Management system. In addition to alumni, stakeholders in our project included **Stella**

Wayianzuvuko (*Principal, Liquidnet Family High School*) and staff at the Career Resource Center, including **Julius Kaboyo** and **Isaac Ouma**.

Technology Infrastructure

The software infrastructure in relation to the alumni management system includes:

Software	Description
PostgreSQL (database)	Postgres is a reliable, open-source database software that supports complex queries, transactions, and JSON data. It's widely used for scalable and flexible data management
Django backend	Django is a high-level, open-source Python web framework that follows the Model-View-Template (MVT) architectural pattern. It helps developers build secure, scalable web applications quickly by providing built-in components for handling data models, user interfaces (templates), and application logic (views). Django includes features like authentication, database ORM (Object-relational mapping - align programming code with database structures), URL routing, and admin interface out of the box, making development efficient and maintainable.
React frontend	React is an open-source JavaScript library for building user interfaces, especially single-page applications. It uses a component-based architecture.
Github (remote repository and collaborative workflow)	GitHub is a web-based platform for version control and collaboration, built around the Git system. It allows developers to host, review, and manage code repositories, track changes, and work together on projects efficiently.
Figma (creating mock-ups and prototyping)	Figma is a design tool used for creating user interfaces, prototypes, and collaborative design projects. It allows multiple users to work together in real time, making it popular for UI/UX design and teamwork.

Technology Management

There is an IT department at ASYV. The head of the department is Deo, and there is also Justin, who is an IT officer and python teacher. ASYV follows the CMU CS Academy to teach python and then expand into HTML, CSS, Javascript to teach web development. Joseph is in charge of the school's learning management system: from software to hardware. He is also responsible for the development and maintenance of the Alumni Management System, which CMU interns from TCinGC have contributed to in the past. There are other staff members who are responsible for collecting and cleaning the alumni data. When teachers and students are facing technical issues, Joseph is usually the person to go to.

Technology Planning

ASYV has long hoped to develop its own learning management system that would handle teaching materials, assignments, attendance, etc. There are many existing options, yet due to how unique ASYV is, most of these options could not fully realize the needs of the organization. Joseph had developed a small management system in PHP in the past few years. Recently, ASYV has adopted AcademicBridge as its learning management system and is currently in the process of testing how it can integrate all departments of school, leap, and parental together. Currently, it is used for grades and attendance.

In 2021 and 2022, ASYV began the national Educational Resilience Program to train teachers in student-centered education and digital teaching technologies.

The long-term goal of the organization is to create one integrated platform to manage all affairs in village lives and school, the current Alumni Management System (AMS) could be regarded as a first step towards this long term goal. ASYV hopes that the development of the AMS could introduce necessary digital infrastructure for future development. The end goal is to create one database that manages all data in the village. The IT department believes that once the organization database has been implemented, it would also make future application development smoother. Our project has taken steps to creating an Integrated Management Information System that can continue to be built upon.

Communication

For general information related to communication, the main language in Rwanda is Kinyarwanda. However, most Rwandans can also speak a bit of English. Currently, we have not experienced any difficulties communicating with the ASYV staff in English. English has been the primary language used for teaching in the school systems since 2008. As for the time zone, Rwanda operates on Central African Time, and does not have daylight savings time.

All ASYV staff have an ASYV google account, but students and alumni do not. Files are generally shared through Google Suites. We have not experienced any difficulties or inefficiencies in regards to communication.

WhatsApp is commonly used for internal communication while emails and video calls are more common for external communication. Communication with families and other stakeholders within Rwanda primarily occurs through phone calls. The New York office handles most of the external communication with donors, visitors, and volunteers. ASYV maintains a website, asyv.org, which serves as a platform for engaging potential donors and visitors. Social media platforms, including

YouTube, Facebook, and Instagram, are actively used by ASYV to raise awareness about the school and gather funds. They can be located on these social media platforms through the following means:

Facebook: [@agahozoshalom](#)

Instagram: [@agahozoshalom](#)

Blog: www.asyv.org/blog

YouTube: www.youtube.com/theasyv

Information Management

Alumni information is critical to the continued success of ASYV. Outcome data can be used to secure funding for ASYV and also assist the CRC (Career Resource Center) in making more informed decisions to support current students at ASYV. The alumni management system also acts as a support network for alumni to keep in touch with each other and to access opportunities such as job openings, reunion events, etc. that they might otherwise not be aware of.

Currently, managing the alumni information is a challenge. Collected information is stored as an Excel spreadsheet to be used for analysis by the administrative team. However, this information might be out of date or inaccurate. Existing methods to collect data consist of using surveys and calling the alumni but this is tedious and time consuming.

The database for the existing Alumni Management System is hosted in Postgres, a relational database management system. The database had been refactored recently by Joseph, into an *integrated management information system (IMIS)* to track not only alumni information but ideally information across all members of ASYV. However, this database design was still missing the ability to track some crucial student information such as college applications and outcomes. The organization and tracking of different roles for each user had room for improvement.

Previously, this refactored database had not been implemented with the existing front-end system.

Improvements to the IMIS are expected to provide immense value to existing and future students of ASYV and ensure that ASYV can efficiently track student outcomes to report and evaluate their programs.

Business Systems

Data on alumni outcomes is a critical motivator for donors and other stakeholders. The alumni management system can help make data collection easier and provide an effective way to assess the data.

II. Making the Alumni Management System Responsive

Motivation

At the start of the project the existing front-end of the AMS website had a working desktop view, but most sections and features broke and became unusable when viewed on a mobile device.

Why it needed to be solved:

- **Popularity of mobile phones:** Most alumni have a mobile phone but may not have access to a laptop or desktop computer. Alumni who only have access to mobile devices could not use the alumni platform.
- **Engagement:** In order to make the AMS as effective as possible, alumni must be engaged with the platform. With a working mobile view, alumni will be able to engage with the platform anytime and anywhere.

By adding responsive functionality to the website, we aimed to make the platform more accessible. A mobile view of the interface allows ASYV to provide support to its alumni regardless of their access to a computer. Providing accessible information on jobs, support on applications, and networking information directly aligns with ASYV’s mission to empower Rwandan youth “to build lives of dignity and contribute to a better world.”

Outcomes

To implement responsive design across the AMS platform, our team conducted a comprehensive front-end refactor, resulting in changes across every major page. Key activities and deliverables included:

- Media query implementation:
 - At widths below 768px, the interface displays a fully mobile-optimized layout.
 - At 768–1024px, the interface maintains a tablet-friendly view, hiding the sidebar and preserving usability.
 - At widths above 1024px, the full desktop layout with sidebar is maintained.
- Navigation redesign:
 - Replaced persistent sidebar navigation with a hamburger menu in the top-right corner for mobile and tablet views.
 - Ensured all routes and links remained accessible and clearly structured across viewports.
- Layout simplification:
 - Converted most pages from two-column layouts to single-column layouts at all screen sizes to improve content flow and component scalability.
 - Updated grid-based pages (e.g., Events and Opportunities) to use a responsive grid that dynamically adjusts the number of columns based on screen size.
- Component restructuring for mobile:
 - Alumni Profile: Tables were restructured into vertically stacked cards with fields listed clearly for small screens.
 - Alumni Directory: The desktop table view was replaced with a simplified, single-column card format for mobile, maintaining all essential data in a digestible layout.
- User testing on mobile devices:
 - Conducted live user testing with alumni using four different mobile phone models.
 - Confirmed that all critical features—viewing profiles, browsing opportunities, editing information, and using the search and filter tools—were functional and accessible.

Outcomes achieved:

- The AMS platform is now fully usable across mobile, tablet, and desktop devices.
- Alumni are able to independently access features on their own phones, including those without computer access.
- The responsive redesign has made information more accessible, improving inclusivity and reducing friction in user engagement.

Indicators:

- Successful user testing on four distinct phone models with no critical bugs or blockers identified.
- Alumni were able to complete common tasks—such as navigating directories and editing their profiles—without external assistance.

The full conversion of all primary pages and components to a mobile-friendly design.

Limitations and Recommendations

Despite the technical success of the responsive redesign, some limitations were observed during user testing:

- Certain features, while functional, caused hesitation among users, indicating room for improvement in UX clarity and interaction design.
- In the staff interface, several administrative functions remain cumbersome on smaller screens. However, given that staff typically access the AMS via desktops, this presents a lower-priority concern.

Future enhancements to optimize UX—particularly for first-time mobile users—could further strengthen platform engagement and accessibility.

III. Redesign of Database and Website Interface

Motivation

The long-term goal for ASYV is to establish a centralized and integrated management information system (MIS) that tracks student outcomes from enrollment through post-graduation. The Alumni Management System (AMS) represents an initial step in this direction, but its original design fell short of meeting the needs of all stakeholders, especially the Career Resource Center (CRC), which depends on accurate and complete data to provide support and evaluate programs.

At the outset of our project, the existing database focused narrowly on alumni records and lacked the flexibility to track critical information such as LEAP program participation, academic performance, and application outcomes. Additionally, the system's interface and underlying data structures were not well-suited to filterable analytics or future expansion. With the frontend still under development and the backend not yet fully launched, we had a key opportunity to redesign the underlying infrastructure in parallel with the user interface—ensuring both functionality and sustainability.

Our team approached this redesign with the goals of increasing platform usability, enabling robust data tracking, and improving the accessibility of meaningful analytics for ASYV staff.

Activities and Outputs

Database Refactor

We fully reorganized and modularized the existing database to enable scalable, flexible data management aligned with user needs. While the total scope of information tracked remained similar, the new schema introduced structural improvements that allow data to be stored, accessed, and presented more effectively.

Key improvements include:

- Expanded staff role tracking with new fields to distinguish CRC, teachers, mamas, and general staff.
- Support for active student tracking, including academic grades, family assignments, and LEAP program participation.
- A graduation status flag to smoothly transition students to alumni within the same system.
- Creation of a Current Student Directory, in parallel with the existing Alumni Directory, to streamline current student management.
- A normalized, modular table structure designed for performance, clarity, and compatibility with frontend filters and analytics.

Admin Data Management Tools

To enable CRC and administrative staff to manage this expanded data more independently, we developed a suite of frontend tools for uploading, editing, and maintaining database records.

Deliverables include:

- Excel template-based import functionality for student related data
- Admin panels for adding and updating colleges, combinations, and LEAP programs.
- Interfaces for managing staff accounts and role assignments.
- Grade entry forms with family assignment functionality.

Staff will be able to manage data in a more streamlined and efficient manner than their current spreadsheet system

Advanced Filtering in Directories

We implemented a new filtering system for both the Alumni and Student Directories. These filters allow users to dynamically narrow down lists based on multiple categories, each with multi-select checkboxes and search functionality.

Available filters include:

- Gender
- Grade
- Family
- Academic Combination
- Industry
- College

While some filters are not applicable to current students, the system uses a shared component and logic to maintain a consistent user experience. These filters significantly enhance the utility of the directory views, allowing CRC staff to quickly locate subsets of users for outreach, reporting, or support.

This feature was only possible due to the modular redesign of the database, which ensured that each filterable property is stored and related in a standardized way.

Staff Dashboard for Analytics

To help the CRC team make better use of alumni data, we designed and implemented a new Staff Dashboard with interactive, filterable visualizations. This tool allows staff to:

- View employment and further education outcomes for alumni by grade and gender.
- Identify the most common colleges, employers, and industries represented in alumni records.
- Filter data dynamically to generate targeted insights.

This dashboard provides ASYV staff with a powerful tool for monitoring trends, informing programming decisions, and preparing reports. Its development was made possible by the normalized data structure and centralized tracking introduced during the refactor.

Alumni Dashboard Redesign

We also redesigned the Alumni Dashboard to better support alumni engagement with the platform. The original dashboard simply replicated the sidebar navigation with static links. We replaced it with dynamic content relevant to alumni, including:

- A preview of upcoming events
- A preview of job opportunities
- A preview of further education opportunities

Each of these content areas have a dedicated page where alumni can explore additional listings. On the staff side, administrators are able to create, manage, and update these listings, ensuring fresh and relevant content for alumni users.

By centralizing career and education content on the dashboard, we aimed to make the AMS more valuable in the day-to-day lives of alumni, thereby increasing platform engagement.

Outcomes and Indicators

Outcomes Achieved:

- The database is now structured to track the full student lifecycle—from enrollment to post-graduation outcomes.
- CRC and administrative staff can manage and update data directly through a user-friendly interface.
- Alumni engagement is supported through personalized dashboards and access to events and opportunities.
- The CRC can now access dynamic, filterable analytics to support advising, reporting, and program development.
- System design supports scalability and ease of future feature expansion.

Indicators:

- A fully revised entity-relationship diagram (ERD) and data dictionary were developed and implemented.
- Student and staff data were successfully uploaded using Excel templates during testing.
- The admin panel was used to populate the system with colleges, LEAP programs, and combinations.
- Alumni and student filters load dynamically and return accurate results across all test cases.
- The staff analytics dashboard displays real-time charts and statistics from the live database.
- Event and opportunity listings were added, browsed, and updated successfully by both alumni and staff.

Limitations

While the redesigned system provides significant improvements in functionality, flexibility, and scalability, several limitations remain:

Data Inaccuracies and Incompleteness:

- Although the new database structure is fully implemented and populated with existing data, the dataset contains known gaps and inconsistencies. These stem from limitations in the original data collection process and the historical absence of a centralized tracking system. As a result, some alumni records are incomplete or outdated.
- To mitigate this, staff can use the newly implemented Manage Data interface to review and correct data manually. Over time, this tool will support the gradual improvement of data accuracy through CRC-led cleanup efforts.
- We have provided data cleaning documentation in Appendix C that lists some of the data inaccuracies we've observed.

Enhancement Opportunities:

- While key functionality such as filtering, events, and opportunity listings are implemented and operational, there are several usability and feature enhancements that remain outstanding. These are not critical for launch but would meaningfully improve user experience and engagement. These suggestions are discussed further in the Recommendations section.

IV. Additional Recommendations

Support Requests for Opportunities

We recommend implementing a backend feature that enables alumni to request CRC support for specific career or further education opportunities directly through the platform. The front end is already partially implemented: each opportunity card includes a placeholder for a “Request CRC Support” button. Once clicked, this action should log a support request in a new database table.

This table would support two key views:

- Alumni View: Alumni can see a list of all their submitted support requests and their statuses.
- Staff View: CRC and other relevant staff can view a consolidated list of support requests by alumni, enabling them to follow up efficiently.

This feature was strongly and consistently requested during user testing with alumni, who emphasized its importance for improving engagement and support access.

Implementing this feature will improve both alumni engagement and staff workflow efficiency:

- **Centralized Communication:** CRC's goal of reducing WhatsApp reliance for alumni support aligns directly with this recommendation. A platform-native support channel reduces fragmentation and data loss while improving accessibility for both alumni and staff.
- **Increased Platform Engagement:** Alumni expressed that this functionality would increase their likelihood of using the platform regularly, especially when applying to jobs, internships, or academic programs.
- **Measurable Impact:** CRCs will be able to track the volume and types of support requests over time, generating actionable data to inform programming, outreach, and resource allocation.

Development steps:

- Create a `SupportRequest` table in the backend database with fields such as: `id`, `user_id`, `opportunity_id`, `status`, `notes`, and `timestamps`.
- Add appropriate API endpoints for creating, updating, and retrieving support requests by alumni and staff.
- Set permissions so that alumni can only view and submit their own requests, and staff can access all records.
- Connect it with the support requests front end tables.

RSVP for Events

We recommend completing the RSVP functionality for events by implementing the necessary backend support. The front end already includes an "RSVP" button on each event card. When completed, clicking this button should log an RSVP in a new database table, allowing:

- Alumni to view a list of events they have RSVP'd to.
- Staff to view the list of alumni who have RSVP'd for each event.

This system would centralize event engagement data, enabling staff to manage attendance more effectively and follow up with alumni who express interest.

This feature will improve alumni participation, event management, and platform stickiness:

- **Better Engagement Tracking:** Knowing who has RSVP'd helps staff personalize outreach and estimate attendance, allowing for better event planning.
- **Higher Alumni Retention:** Alumni are more likely to return to a platform where they can actively participate in events and track their involvement.
- **Organizational Efficiency:** Staff no longer need to rely on external forms or informal messages for RSVP tracking. Everything can happen directly on the platform.

Development steps:

- The structure for this feature already exists in both the front and back end
- Finalizing and testing the connection would complete it

Resume Upload

We recommend allowing alumni to upload their resumes directly to their profiles. This would involve adding a file upload field to the alumni profile page and storing the uploaded document in

the backend. Staff (particularly CRC) would be able to view resumes while reviewing alumni data, and alumni could update their resume over time.

This feature was consistently requested by both alumni and CRC staff during user testing and project discussions. CRC regularly supports alumni with job applications and further education — having resumes easily accessible will greatly enhance this process.

Adding a resume upload feature would create a more complete and supportive alumni profile, directly aiding CRC's core responsibilities:

- Streamlined Support: CRC can access and review resumes without having to request them over WhatsApp or email, saving time for both alumni and staff.
- Improved Context: When CRC responds to opportunity support requests (above), having a resume on file provides context about the alumni's experience and goals.
- Centralized Records: It supports ASYV's broader goal of reducing reliance on fragmented communication and creating a centralized alumni engagement platform.

Development steps:

- Add a resume field to the model
- Create API endpoints for uploading, replacing, and retrieving resumes.
- Add front end abilities to add and view resume

This is a lightweight yet highly impactful feature that supports CRC's workflow and improves the platform's value to alumni seeking opportunities.

Add More Dropdowns and Data Validation to Forms

We recommend enhancing the data entry forms on the platform by replacing more free-text fields with dropdowns and adding additional validation checks. This will preserve data integrity and reduce inconsistencies that often arise from manual entry. While the system currently uses dropdowns for some repeated fields (such as combinations and industry), others are still entered as text and are vulnerable to typos or inconsistent labeling.

Improving form validation and standardization is a natural next step now that the core database and user interface have been rebuilt.

This recommendation directly supports data accuracy and long-term usability of the system:

- Cleaner data for analytics and filtering: With standardized dropdown entries, charts, filters, and exports will return more reliable results.
- Less manual cleanup needed by staff: Avoids issues like "IT," "it," and "Information Technology" being entered as separate values.
- Scalability: As the dataset grows, dropdowns and validation help maintain order and consistency across a larger system.
- Improved user experience: Dropdowns reduce friction and guesswork for staff and alumni entering data.

Development steps:

- Audit current forms looking for necessary validations and fields where a dropdown would be better
- Add validation and dropdown logic

This recommendation builds directly on the foundations laid during the consulting project and will help ensure the long-term cleanliness and usability of the alumni management system.

Development of an Alumni Connection Network

We recommend that ASYV support the development of a dedicated alumni connection platform—a social-media-like space where alumni can reach out to each other, build connections, and directly message one another. Both alumni and CRC staff consistently requested this feature throughout our user interviews and testing sessions. It would also help move CRC support conversations off WhatsApp and consolidate interactions on a centralized ASYV platform.

We determined that building such a platform was outside the scope of this consulting engagement due to its complexity and the time required. However, we recently connected with a student who is developing a mentorship and alumni networking platform as a personal project. The project closely mirrors the CRC's vision for alumni connectivity. We have introduced this student to Joseph and strongly recommend that ASYV explore a collaboration to help bring this solution to life.

Supporting an internal effort like this leverages local talent, meets a clear need identified by the CRC and alumni, and allows ASYV to pilot a connection platform in a way that aligns with its values and capacity.

All resources for these recommendations are available. The features can be implemented with Joseph's expertise and potentially help from IT interns at ASYV.

About the Consultants

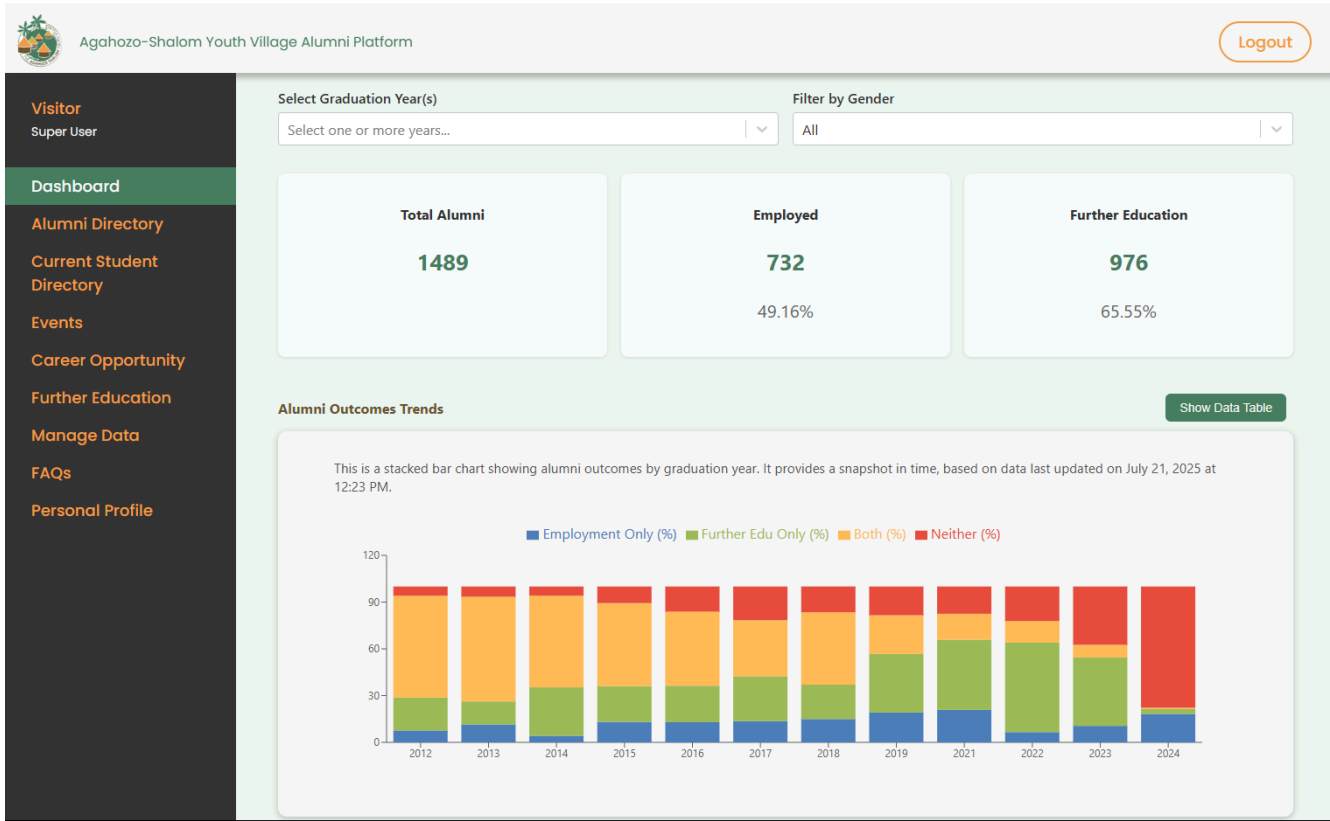
Jiatong Li graduated Carnegie Mellon University in May 2025 with a Bachelors in Information Systems with an additional major of Human-Computer Interaction. In the fall she will be starting her Masters in Healthcare Analytics at Carnegie Mellon University.

Kajal Mehta graduated Carnegie Mellon University in May 2025 with a Bachelors in Information Systems and a minor Human-Computer Interaction. In the fall she will be starting as a product manager at Capital One.

Appendix A.
Screenshots of interface

Staff View on Desktop

Staff Dashboard





Visitor
Super User

Dashboard

Alumni Directory

Current Student
Directory

Events

Career Opportunity

Further Education

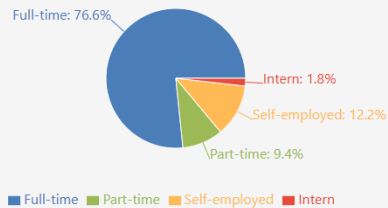
Manage Data

FAQs

Personal Profile

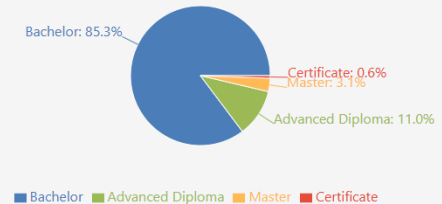
Employment Status Distribution

The distribution of type of employment for alumni who are employed.



Degree Level Distribution

The distribution of degree levels pursued by alumni, such as Bachelor's, Master's, or Ph.D for alumni who pursued further education.



Colleges Attended by Country

A breakdown of colleges attended by alumni, grouped by country. The size of each group reflects the number of alumni who studied in each country. Click on a country to view colleges.



Areas of Study

This list shows the most common academic disciplines alumni pursued.

Hospitality Management: 91
Finance: 77
Economic: 48
Accounting: 47
Information System: 43
Marketing: 40
Business Management and Entrepreneurship: 40
Social Science: 38
Information Technology: 37
Agriculture: 26
Law: 22



Visitor
Super User

Dashboard

Alumni Directory

Current Student
Directory

Events

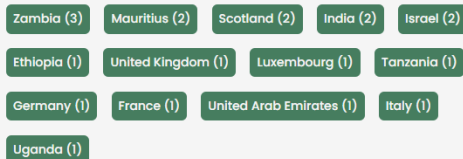
Career Opportunity

Further Education

Manage Data

FAQs

Personal Profile



Business Management and Entrepreneurship: 40
Social Science: 38
Information Technology: 37
Agriculture: 26
Law: 22
Journalism and Communication: 18
International Hotel Management: 16
Computer Science: 14
Business Information and Technology: 13
General Nursing: 13
Photography and Movie Editing: 12

Industry Distribution

Shows the distribution of industries alumni are employed in.

Business and Management: 280
Education and Training: 118
Dining and Hospitality Services: 116
Information Technology and Software Development: 65
Healthcare and Medical Services: 40
Construction and Engineering: 23
Agriculture, Forestry, and Fishing: 21
Finance and Banking: 14
Art, Design, and Performance: 13
Media and Communication: 13
Security and Law Enforcement: 11
Others / Not Specified: 6
Energy and Utilities: 3
Government and Public Service: 2
Sports and Recreation: 2
Trade and Skilled Labor: 2
Beauty and Personal Care: 1

Top 10 Employers

Lists the companies that employ the highest number of alumni.

Agahozo-Shalom Youth Village: 54 alumni
Self-employed: 13 alumni
Local government: 13 alumni
Rwanda Nation Police: 13 alumni
Envision Rwanda: 10 alumni
Rwanda Defence Forces: 6 alumni
F&D company: 5 alumni
Clothes Shop: 4 alumni
Rwanda Investigation Bureau: 4 alumni
Rwandair: 4 alumni

Events



Agahozo-Shalom Youth Village Alumni Platform

Logout

Visitor
Super User

Dashboard

Alumni Directory

Current Student
Directory

Events

Career Opportunity

Further Education

Manage Data

FAQs

Personal Profile

Add Event



Alumni Reunion

2025-08-04

ASYV

DeleteEdit



Alumni Hike

2025-08-25

ASYV

DeleteEdit

Calendar

Photos

Opportunities (Career and Education)



Agahozo-Shalom Youth Village Alumni Platform

Logout

Visitor
Super User

Dashboard

Alumni Directory

Current Student
Directory

Events

Career Opportunity

Further Education

Manage Data

FAQs

Personal Profile

Full Time

Part Time

Internship

Volunteer

Create Draft

Click on an opportunity to view more details. Click on the button to view the application link.

User Tester

Apply by: 2025-07-26

ASYV

[TEST] Help test our new alumni platform.

Hide

Edit

Sample job

Apply by: 2025-08-01

ASYV

[TEST] This is a sample job to give a feel for what can be explored on the platform

Hide

Edit

https://asyv.ac.rw/career_opportunity_staff

Managing Data Page



Agahozo-Shalom Youth Village Alumni Platform

Logout

Visitor
Super User

Dashboard

Alumni Directory

Current Student
Directory

Events

Career Opportunity

Further Education

Manage Data

FAQs

Personal Profile

Manage Data

Colleges

AddView

Leaps

AddView

Combinations

AddView

Grade & Families

AddView

Students

Add

Staff Account

AddView

Leaps

[Add](#)[Close List](#)

3D Modeling — science center

[Edit](#)[Delete](#)

Carpentry — art center

[Edit](#)[Delete](#)

Computer literacy — professional

[Edit](#)[Delete](#)

Drawing — art center

[Edit](#)[Delete](#)

Drums — art center

[Edit](#)[Delete](#)

Leaps

[Close Form](#)[View](#)

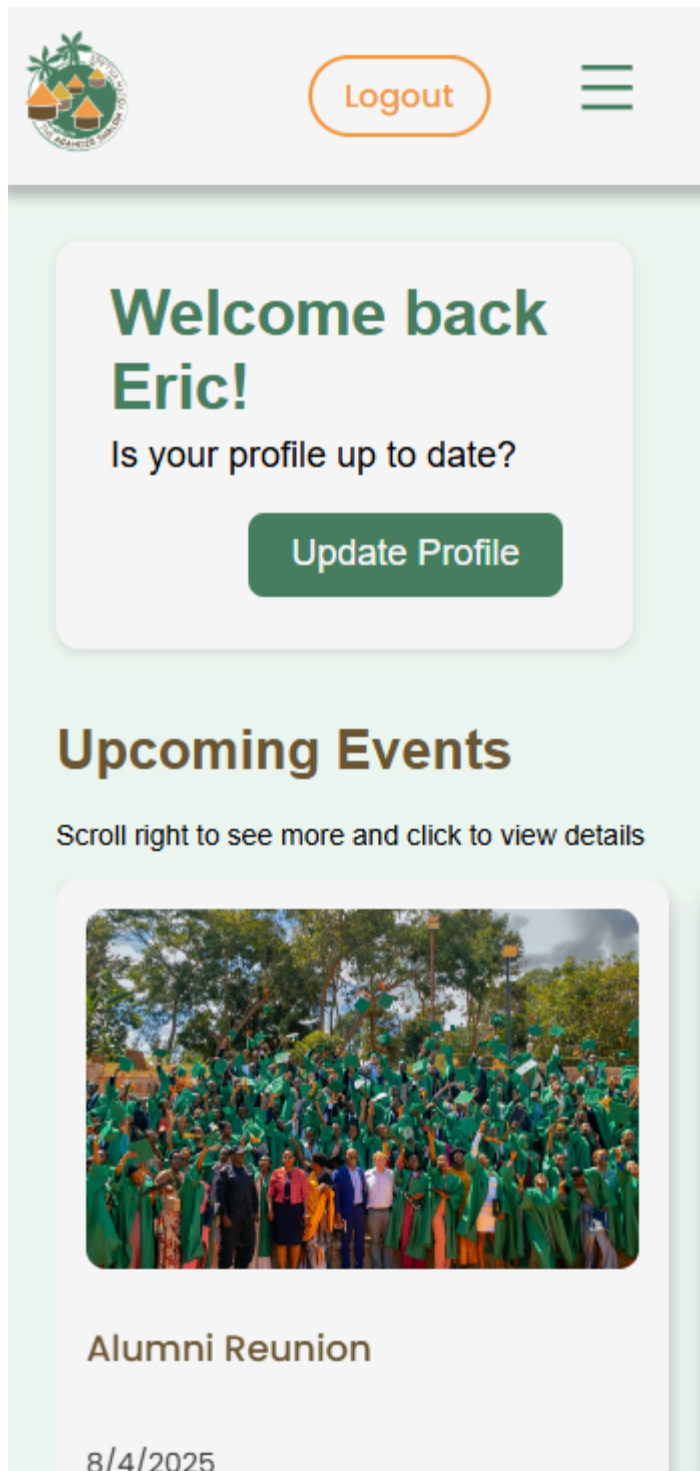
Name (ep):

Category:

[Add](#)[View Existing Entries →](#)

Alumni View on Mobile

Alumni Dashboard



Events



Logout

Calendar

Photos



Alumni Reunion

2025-08-04

ASYV



Opportunities (Career and Education)



Logout

Full Time

Click on an opportunity to view more details.
Click on the button to view the application link.

User Tester

Apply by: 2025-07-26

ASYV

[TEST] Help test our new alumni platform.

Apply

Sample job

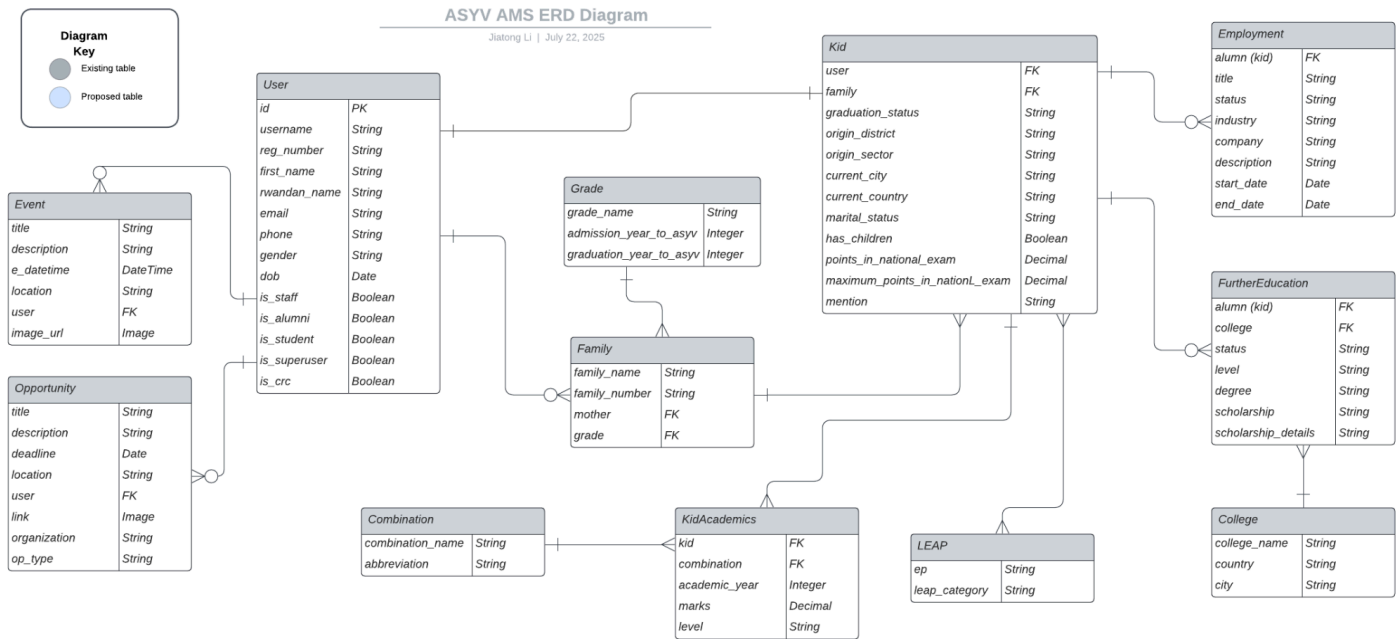
Apply by: 2025-08-01

ASYV

[TEST] This is a sample job to give a feel for what can be explored on the platform

Appendix B.

Refactored ERD



Appendix C.

Staff User guide -  *Documentation - User Guide (STAFF)*

Developer User guide -  *Documentation - Developer Guide*

Excel Download Templates -  *Data Upload Templates*

Data Cleaning Documentation -  *Data Cleaning Documentation*